

Volume I Mathematical Logic And Proof Reference

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Propositional Logic

0.1 Model-Theoretic View

0.2 Axiom Schemas

0.3 Notation and Conventions

0.4 Syntax

Definition 0.4.1 (Truth Value). The set of truth values is

$$\mathbb{B} := \{\text{False}, \text{True}\}.$$

Some sources identify this set with $\{0, 1\}$, with 0 read as false and 1 read as true.

Definition 0.4.2 (Propositional Language). A propositional language consists of:

1. a set **Prop** of propositional variables;
2. logical connectives;
3. punctuation symbols such as parentheses;
4. recursive formation rules specifying the well-formed formulas.

Definition 0.4.3 (Propositional Variable). A propositional variable is an atomic symbol intended to stand for a proposition. A standard supply of variables is

$$\text{Prop} = \{P_1, P_2, P_3, \dots\}.$$

Informally, variables are often written $P, Q, R, S, \dots$

Definition 0.4.4 (Logical Connective). The standard primitive propositional connectives are

$$\neg, \quad \wedge, \quad \vee, \quad \rightarrow, \quad \leftrightarrow.$$

The connective $\neg$ is unary. The connectives $\wedge, \vee, \rightarrow, \leftrightarrow$ are binary.

Symbol	Name	Arity	Formation pattern	Reading
$\neg$	negation	unary	$\neg\varphi$	not φ
$\wedge$	conjunction	binary	$(\varphi \wedge \psi)$	φ and ψ
$\vee$	disjunction	binary	$(\varphi \vee \psi)$	φ or ψ
$\rightarrow$	conditional	binary	$(\varphi \rightarrow \psi)$	if φ , then ψ
$\leftrightarrow$	biconditional	binary	$(\varphi \leftrightarrow \psi)$	φ iff ψ

Definition 0.4.5 (Well-Formed Formula). The set **WFF** of well-formed formulas is the smallest set of strings satisfying the following formation rules:

1. Every propositional variable $P \in \text{Prop}$ belongs to WFF.
2. If $\varphi \in \text{WFF}$, then $\neg\varphi \in \text{WFF}$.
3. If $\varphi, \psi \in \text{WFF}$ and $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then $(\varphi \circ \psi) \in \text{WFF}$.
4. No string is a well-formed formula unless it is obtained by finitely many applications of the previous clauses.

Lemma 0.4.6 (Closure Under Formation). *The set WFF is closed under the propositional formation rules. That is, if $\varphi \in \text{WFF}$, then*

$$\neg\varphi \in \text{WFF}.$$

If $\varphi, \psi \in \text{WFF}$ and

$$\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\},$$

then

$$(\varphi \circ \psi) \in \text{WFF}.$$

Go to proof.

Theorem 0.4.7 (Minimality of Well-Formed Formulas). *Let S be a set of strings over the propositional alphabet. Suppose:*

1. $\text{Prop} \subseteq S$;
2. if $\varphi \in S$, then $\neg\varphi \in S$;
3. if $\varphi, \psi \in S$ and $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then $(\varphi \circ \psi) \in S$.

Then

$$\text{WFF} \subseteq S.$$

Go to proof.

Definition 0.4.8 (Atomic Formula). An atomic formula is a well-formed formula that is a propositional variable.

Definition 0.4.9 (Molecular Formula). A molecular formula is a well-formed formula that is not atomic. Equivalently, it is a well-formed formula formed using at least one logical connective.

Definition 0.4.10 (Main Connective). The main connective of a molecular formula is the connective introduced at the final formation step:

1. for $\neg\varphi$, the main connective is $\neg$;
2. for $(\varphi \circ \psi)$, the main connective is $\circ$.

Atomic formulas have no main connective.

Theorem 0.4.11 (Structural Induction for Propositional Formulas). *Let $A(\varphi)$ be a property of well-formed formulas. Suppose:*

1. $A(P)$ holds for every $P \in \text{Prop}$;
2. if $A(\varphi)$ holds, then $A(\neg\varphi)$ holds;
3. if $A(\varphi)$ and $A(\psi)$ hold, then $A((\varphi \circ \psi))$ holds for every $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$.

Then $A(\theta)$ holds for every $\theta \in \text{WFF}$.

Go to proof.

Theorem 0.4.12 (Structural Recursion for Propositional Formulas). *To define a function F on WFF, it suffices to specify:*

1. $F(P)$ for every $P \in \text{Prop}$;
2. $F(\neg\varphi)$ in terms of $F(\varphi)$;
3. $F((\varphi \circ \psi))$ in terms of $F(\varphi)$, $F(\psi)$, and $\circ$.

When these clauses respect the formation rules, they determine a unique function on WFF.

Go to proof.

Lemma 0.4.13 (Constructor Disjointness for Propositional Formulas). *The three outer formation cases for propositional formulas are disjoint:*

1. no propositional variable is a negation formula;
2. no propositional variable is a binary formula;
3. no negation formula is a binary formula.

Go to proof.

Lemma 0.4.14 (Constructor Injectivity for Propositional Formulas). *The propositional formula constructors are injective:*

1. if $\neg\varphi = \neg\psi$, then $\varphi = \psi$;
2. if $(\varphi \circ \psi) = (\alpha \star \beta)$, then $\varphi = \alpha$, $\psi = \beta$, and $\circ = \star$.

Go to proof.

Theorem 0.4.15 (Unique Decomposition for Propositional Formulas). *Every $\varphi \in \text{WFF}$ has exactly one outermost form:*

1. $\varphi = P$ for a unique $P \in \text{Prop}$;
2. $\varphi = \neg\psi$ for a unique $\psi \in \text{WFF}$;
3. $\varphi = (\psi \circ \chi)$ for unique $\psi, \chi \in \text{WFF}$ and a unique binary connective $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$.

Go to proof.

Definition 0.4.16 (Parse Tree). The parse tree, or syntax tree, of a well-formed formula is the tree representation of its recursive construction:

1. a propositional variable is represented by a single leaf;
2. a formula $\neg\varphi$ is represented by a root labeled $\neg$ with one child, the parse tree of φ ;
3. a formula $(\varphi \circ \psi)$ is represented by a root labeled $\circ$ with two children, the parse trees of φ and ψ .

Theorem 0.4.17 (Unique Syntax Tree for Propositional Formulas). *Every propositional well-formed formula has a unique parse tree.*

Go to proof.

Definition 0.4.18 (Variable Set of a Formula). The variable set $\text{Var}(\varphi)$ of a formula φ is defined recursively:

1. if $\varphi = P \in \text{Prop}$, then $\text{Var}(\varphi) = \{P\}$;
2. if $\varphi = \neg\psi$, then $\text{Var}(\varphi) = \text{Var}(\psi)$;

3. if $\varphi = (\psi \circ \chi)$, then

$$\text{Var}(\varphi) = \text{Var}(\psi) \cup \text{Var}(\chi).$$

Definition 0.4.19 (Subformula). The set $\text{Sub}(\varphi)$ of subformulas of a formula φ is defined recursively:

1. if φ is atomic, then $\text{Sub}(\varphi) = \{\varphi\}$;

2. if $\varphi = \neg\psi$, then

$$\text{Sub}(\varphi) = \{\varphi\} \cup \text{Sub}(\psi);$$

3. if $\varphi = (\psi \circ \chi)$, then

$$\text{Sub}(\varphi) = \{\varphi\} \cup \text{Sub}(\psi) \cup \text{Sub}(\chi).$$

Definition 0.4.20 (Proper Subformula). Let $\varphi \in \text{WFF}$. A *proper subformula* of φ is a formula ψ such that

$$\psi \in \text{Sub}(\varphi) \quad \text{and} \quad \psi \neq \varphi.$$

Definition 0.4.21 (Formula Depth). The depth of a formula φ , denoted $\text{depth}(\varphi)$, is defined recursively:

1. if φ is atomic, then $\text{depth}(\varphi) = 0$;

2. if $\varphi = \neg\psi$, then $\text{depth}(\varphi) = \text{depth}(\psi) + 1$;

3. if $\varphi = (\psi \circ \chi)$, then

$$\text{depth}(\varphi) = \max\{\text{depth}(\psi), \text{depth}(\chi)\} + 1.$$

Definition 0.4.22 (Connective Count). The connective count of a formula φ , denoted $\text{conn}(\varphi)$, is defined recursively:

1. if φ is atomic, then $\text{conn}(\varphi) = 0$;

2. if $\varphi = \neg\psi$, then $\text{conn}(\varphi) = \text{conn}(\psi) + 1$;

3. if $\varphi = (\psi \circ \chi)$, then

$$\text{conn}(\varphi) = \text{conn}(\psi) + \text{conn}(\chi) + 1.$$

Lemma 0.4.23 (Proper Subformula Depth). *If ψ is a proper subformula of φ , then*

$$\text{depth}(\psi) < \text{depth}(\varphi).$$

Go to proof.

Lemma 0.4.24 (Finiteness of Variables in a Formula). *For every $\varphi \in \text{WFF}$, the set $\text{Var}(\varphi)$ is finite.*

Go to proof.

Lemma 0.4.25 (Finiteness of Subformulas). *For every $\varphi \in \text{WFF}$, the set $\text{Sub}(\varphi)$ is finite.*

Go to proof.

0.5 Semantics

Truth Assignments

Definition 0.5.1 (Truth Assignment). A *truth assignment* is a function

$$v : \text{Prop} \rightarrow \mathbb{B}.$$

For each propositional variable $P \in \text{Prop}$, the value $v(P)$ is the truth value assigned to P .

Definition 0.5.2 (Domain of a Truth Assignment). The *domain* of a truth assignment

$$v : \text{Prop} \rightarrow \mathbb{B}$$

is the set Prop of propositional variables.

Extension of a Truth Assignment

Definition 0.5.3 (Extension of a Truth Assignment to Formulas). Let

$$v : \text{Prop} \rightarrow \mathbb{B}$$

be a truth assignment. The *extension* of v to well-formed formulas is the function

$$\hat{v} : \text{WFF} \rightarrow \mathbb{B}$$

defined recursively as follows:

$$\begin{aligned} \hat{v}(P) &= v(P) \quad \text{for every } P \in \text{Prop}, \\ \hat{v}(\neg\varphi) &= \text{True} \iff \hat{v}(\varphi) = \text{False}, \\ \hat{v}(\varphi \wedge \psi) &= \text{True} \iff \hat{v}(\varphi) = \text{True} \text{ and } \hat{v}(\psi) = \text{True}, \\ \hat{v}(\varphi \vee \psi) &= \text{True} \iff \hat{v}(\varphi) = \text{True} \text{ or } \hat{v}(\psi) = \text{True}, \\ \hat{v}(\varphi \rightarrow \psi) &= \text{False} \iff \hat{v}(\varphi) = \text{True} \text{ and } \hat{v}(\psi) = \text{False}, \end{aligned}$$

and

$$\hat{v}(\varphi \leftrightarrow \psi) = \text{True} \iff \hat{v}(\varphi) = \hat{v}(\psi).$$

Theorem 0.5.4 (Unique Extension of a Truth Assignment). *Every truth assignment*

$$v : \text{Prop} \rightarrow \mathbb{B}$$

has a unique extension

$$\hat{v} : \text{WFF} \rightarrow \mathbb{B}$$

satisfying the recursive truth clauses for propositional variables and the primitive connectives.

Go to proof.

Definition 0.5.5 (Satisfaction of a Formula). Let v be a truth assignment and let $\varphi \in \text{WFF}$. We say that v satisfies φ , written

$$v \models \varphi,$$

if

$$\hat{v}(\varphi) = \text{True}.$$

Definition 0.5.6 (Satisfaction of a Set of Formulas). Let $\Gamma \subseteq \text{WFF}$. A truth assignment v satisfies Γ , written

$$v \models \Gamma,$$

if $v \models \gamma$ for every $\gamma \in \Gamma$.

Definition 0.5.7 (Truth Table). Let $\varphi \in \text{WFF}$ and suppose the variables occurring in φ are among $P_1, \dots, P_n$. A truth table for φ is a finite table with one row for each assignment

$$a : \{P_1, \dots, P_n\} \rightarrow \mathbb{B}$$

and a final column recording the value of $\hat{v}(\varphi)$ for any truth assignment v extending a .

Definition 0.5.8 (Tautology). A formula $\varphi \in \text{WFF}$ is a tautology, written

$$\models \varphi,$$

if every truth assignment satisfies φ .

Definition 0.5.9 (Contradiction). A formula $\varphi \in \text{WFF}$ is a contradiction if no truth assignment satisfies φ .

Definition 0.5.10 (Satisfiable Formula). A formula $\varphi \in \text{WFF}$ is satisfiable if at least one truth assignment satisfies it.

Definition 0.5.11 (Contingency). A formula $\varphi \in \text{WFF}$ is a contingency if it is satisfiable but not a tautology.

Definition 0.5.12 (Logical Consequence). Let $\Gamma \subseteq \text{WFF}$ and $\varphi \in \text{WFF}$. The formula φ is a logical consequence of Γ , written

$$\Gamma \models \varphi,$$

if every truth assignment satisfying Γ also satisfies φ .

Definition 0.5.13 (Logical Equivalence). Formulas $\varphi, \psi \in \text{WFF}$ are logically equivalent, written

$$\varphi \equiv \psi,$$

if they have the same truth value under every truth assignment.

Lemma 0.5.14 (Coincidence Lemma for Propositional Logic). *Let $v, w : \text{Prop} \rightarrow \mathbb{B}$ be truth assignments. If v and w agree on every propositional variable occurring in φ , then they assign the same truth value to φ :*

$$v|_{\text{Var}(\varphi)} = w|_{\text{Var}(\varphi)} \implies \widehat{v}(\varphi) = \widehat{w}(\varphi).$$

Go to proof.

Theorem 0.5.15 (Finite Truth-Table Theorem). *Every propositional formula has a finite truth table. More precisely, if $\text{Var}(\varphi)$ has n elements, then φ has a truth table with 2^n rows.*

Go to proof.

Definition 0.5.16 (Unary Boolean Function). A unary Boolean function is a function

$$f : \mathbb{B} \rightarrow \mathbb{B}.$$

Definition 0.5.17 (Binary Boolean Function). A binary Boolean function is a function

$$f : \mathbb{B}^2 \rightarrow \mathbb{B},$$

equivalently a function

$$f : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}.$$

Definition 0.5.18 (Boolean Function). An n -ary Boolean function is a function

$$f : \mathbb{B}^n \rightarrow \mathbb{B}.$$

Definition 0.5.19 (Truth Function Induced by a Formula). Let $\varphi \in \text{WFF}$, and suppose $\text{Var}(\varphi) \subseteq \{P_1, \dots, P_n\}$. The truth function induced by φ is the Boolean function

$$f_\varphi : \mathbb{B}^n \rightarrow \mathbb{B}$$

defined by

$$f_\varphi(b_1, \dots, b_n) = \widehat{v}(\varphi),$$

where $v(P_i) = b_i$ for each i .

Theorem 0.5.20 (Counting Boolean Functions). *There are exactly*

$$2^{2^n}$$

Boolean functions $\mathbb{B}^n \rightarrow \mathbb{B}$.

Go to proof.

0.6 Algebra of Propositions

Definition 0.6.1 (Propositional Equivalence Law). A propositional equivalence law is a valid schema of the form

$$\varphi \equiv \psi,$$

with $\varphi, \psi \in \text{WFF}$. Each instance of the schema says that the two formulas have the same truth value under every truth assignment.

Theorem 0.6.2 (Logical Equivalence is an Equivalence Relation). *Logical equivalence is reflexive, symmetric, and transitive on WFF:*

$$\varphi \equiv \varphi, \quad \varphi \equiv \psi \implies \psi \equiv \varphi,$$

and

$$(\varphi \equiv \psi \wedge \psi \equiv \chi) \implies \varphi \equiv \chi.$$

Go to proof.

Theorem 0.6.3 (Basic Boolean Equivalence Laws). *For all formulas $\varphi, \psi, \chi \in \text{WFF}$, the following logical equivalences hold.*

Commutativity:

$$\varphi \wedge \psi \equiv \psi \wedge \varphi, \quad \varphi \vee \psi \equiv \psi \vee \varphi.$$

Associativity:

$$(\varphi \wedge \psi) \wedge \chi \equiv \varphi \wedge (\psi \wedge \chi), \quad (\varphi \vee \psi) \vee \chi \equiv \varphi \vee (\psi \vee \chi).$$

Idempotence:

$$\varphi \wedge \varphi \equiv \varphi, \quad \varphi \vee \varphi \equiv \varphi.$$

Double negation:

$$\neg\neg\varphi \equiv \varphi.$$

Go to proof.

Theorem 0.6.4 (Identity and Domination Laws). *Let $\top$ be any tautological formula and let $\perp$ be any contradictory formula. Then for every $\varphi \in \text{WFF}$:*

$$\varphi \wedge \top \equiv \varphi, \quad \varphi \vee \perp \equiv \varphi,$$

and

$$\varphi \wedge \perp \equiv \perp, \quad \varphi \vee \top \equiv \top.$$

Go to proof.

Theorem 0.6.5 (De Morgan Laws for Propositional Logic). *For all $\varphi, \psi \in \text{WFF}$,*

$$\neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi,$$

and

$$\neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi.$$

Go to proof.

Theorem 0.6.6 (Distributive Laws for Propositional Logic). *For all $\varphi, \psi, \chi \in \text{WFF}$,*

$$\varphi \wedge (\psi \vee \chi) \equiv (\varphi \wedge \psi) \vee (\varphi \wedge \chi),$$

and

$$\varphi \vee (\psi \wedge \chi) \equiv (\varphi \vee \psi) \wedge (\varphi \vee \chi).$$

Go to proof.

Theorem 0.6.7 (Absorption Laws). *For all $\varphi, \psi \in \text{WFF}$,*

$$\varphi \wedge (\varphi \vee \psi) \equiv \varphi,$$

and

$$\varphi \vee (\varphi \wedge \psi) \equiv \varphi.$$

Go to proof.

Theorem 0.6.8 (Conditional Equivalence Laws). *For all $\varphi, \psi \in \text{WFF}$,*

$$\varphi \rightarrow \psi \equiv \neg\varphi \vee \psi,$$

and

$$\neg(\varphi \rightarrow \psi) \equiv \varphi \wedge \neg\psi.$$

Go to proof.

Theorem 0.6.9 (Biconditional Equivalence Laws). *For all $\varphi, \psi \in \text{WFF}$,*

$$\varphi \leftrightarrow \psi \equiv (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi),$$

and

$$\varphi \leftrightarrow \psi \equiv (\varphi \wedge \psi) \vee (\neg\varphi \wedge \neg\psi).$$

Go to proof.

Definition 0.6.10 (Formula Context). A formula context $C[-]$ is a well-formed formula with one distinguished placeholder occurrence $[-]$. For each well-formed formula φ , the expression $C[\varphi]$ denotes the formula obtained by replacing the placeholder with φ .

Theorem 0.6.11 (Replacement of Logical Equivalents). *Replacing logically equivalent formulas inside a formula context preserves logical equivalence:*

$$\varphi \equiv \psi \implies C[\varphi] \equiv C[\psi].$$

Go to proof.

0.7 Normal Forms

Orientation

Definition 0.7.1 (Literal). A *literal* is either a propositional variable P or the negation $\neg P$ of a propositional variable P .

Definition 0.7.2 (Positive Literal). A *positive literal* is a literal of the form P , where P is a propositional variable.

Definition 0.7.3 (Negative Literal). A *negative literal* is a literal of the form $\neg P$, where P is a propositional variable.

Definition 0.7.4 (Complementary Literals). Two literals are *complementary* if they are P and $\neg P$ for the same propositional variable P .

Definition 0.7.5 (Clause). A *clause* is a finite disjunction of literals.

Definition 0.7.6 (Term). A *term*, also called a *conjunction term*, is a finite conjunction of literals.

Definition 0.7.7 (Empty Clause). The *empty clause* is the empty disjunction of literals. Semantically, it is read as false.

Definition 0.7.8 (Empty Conjunction). The *empty conjunction* is the conjunction over no literals. Semantically, it is read as true.

Definition 0.7.9 (Negation Normal Form). A formula is in *negation normal form* if it uses only propositional variables, negated propositional variables, conjunction, and disjunction, and every negation occurs immediately before a propositional variable.

Definition 0.7.10 (NNF Conversion Procedure). The *NNF conversion procedure* converts a propositional formula to an equivalent formula in negation normal form by the following steps:

1. eliminate biconditionals using biconditional equivalence laws;
2. eliminate conditionals using conditional equivalence laws;
3. eliminate double negations;
4. push negations inward using De Morgan laws.

Theorem 0.7.11 (Existence of Negation Normal Form). *For every formula $\varphi \in \text{WFF}$, there exists a formula $\psi \in \text{WFF}$ such that ψ is in negation normal form and*

$$\varphi \equiv \psi.$$

Go to proof.

Definition 0.7.12 (Elementary Conjunction). An *elementary conjunction* is a finite conjunction of **literals**. That is, it is a **term** of the form

$$L_1 \wedge L_2 \wedge \cdots \wedge L_k,$$

where each L_i is a literal. The one-literal case is permitted.

Definition 0.7.13 (Minterm). Let $P_1, \dots, P_n$ be a finite list of propositional variables. A *minterm* relative to $P_1, \dots, P_n$ is an **elementary conjunction** that contains exactly one of P_i or $\neg P_i$ for each $i = 1, \dots, n$.

Definition 0.7.14 (Disjunctive Normal Form). A formula is in *disjunctive normal form*, abbreviated *DNF*, when it is a finite disjunction of **elementary conjunctions**. Equivalently, it has the shape

$$C_1 \vee C_2 \vee \cdots \vee C_m,$$

where each C_j is an elementary conjunction.

Theorem 0.7.15 (Existence of Disjunctive Normal Form). *For every formula $\varphi \in \text{WFF}$, there exists a formula $\psi \in \text{WFF}$ such that ψ is in **disjunctive normal form** and*

$$\varphi \equiv \psi.$$

Go to proof.

Theorem 0.7.16 (DNF Satisfiability Criterion). *Let ψ be a formula in **disjunctive normal form**. Then ψ is **satisfiable** if and only if at least one conjunction term in ψ contains no **complementary pair of literals**. Go to proof.*

Definition 0.7.17 (Elementary Disjunction). An *elementary disjunction* is a finite disjunction of **literals**. That is, it is a **clause** of the form

$$L_1 \vee L_2 \vee \cdots \vee L_k,$$

where each L_i is a literal. The one-literal case is permitted.

Definition 0.7.18 (Maxterm). Let $P_1, \dots, P_n$ be a finite list of propositional variables. A *maxterm* relative to $P_1, \dots, P_n$ is an **elementary disjunction** that contains exactly one of P_i or $\neg P_i$ for each $i = 1, \dots, n$.

Definition 0.7.19 (Conjunctive Normal Form). A formula is in *conjunctive normal form*, abbreviated *CNF*, when it is a finite conjunction of *clauses*. Equivalently, it has the shape

$$C_1 \wedge C_2 \wedge \cdots \wedge C_m,$$

where each C_j is a clause.

Theorem 0.7.20 (Existence of Conjunctive Normal Form). *For every formula $\varphi \in \text{WFF}$, there exists a formula $\psi \in \text{WFF}$ such that ψ is in *conjunctive normal form* and*

$$\varphi \equiv \psi.$$

Go to proof.

Definition 0.7.21 (Truth-Table Row Conjunction). Fix a finite variable list $P_1, \dots, P_n$ and a row r of the *truth table* for those variables. The *truth-table row conjunction* for r is the conjunction

$$L_1 \wedge L_2 \wedge \cdots \wedge L_n,$$

where

$$L_i = \begin{cases} P_i, & \text{if row } r \text{ makes } P_i \text{ true,} \\ \neg P_i, & \text{if row } r \text{ makes } P_i \text{ false.} \end{cases}$$

Definition 0.7.22 (Truth-Table Row Clause). Fix a finite variable list $P_1, \dots, P_n$ and a row r of the *truth table* for those variables. The *truth-table row clause* for r is the clause

$$M_1 \vee M_2 \vee \cdots \vee M_n,$$

where

$$M_i = \begin{cases} P_i, & \text{if row } r \text{ makes } P_i \text{ false,} \\ \neg P_i, & \text{if row } r \text{ makes } P_i \text{ true.} \end{cases}$$

Definition 0.7.23 (Canonical Disjunctive Normal Form). Let $\varphi \in \text{WFF}$, and fix an ordering $P_1, \dots, P_n$ of the finite variable set of φ . The *canonical disjunctive normal form* of φ relative to this ordering is the disjunction of the *truth-table row conjunctions* corresponding to the rows on which φ is true.

Definition 0.7.24 (Canonical Conjunctive Normal Form). Let $\varphi \in \text{WFF}$, and fix an ordering $P_1, \dots, P_n$ of the finite variable set of φ . The *canonical conjunctive normal form* of φ relative to this ordering is the conjunction of the *truth-table row clauses* corresponding to the rows on which φ is false.

Theorem 0.7.25 (Canonical Disjunctive Normal Form Theorem). *Every formula $\varphi \in \text{WFF}$ is *logically equivalent* to its *canonical disjunctive normal form* relative to a fixed ordering of its finitely many variables.*

$$\varphi \equiv \text{CDNF}_{P_1, \dots, P_n}(\varphi).$$

Go to proof.

Theorem 0.7.26 (Canonical Conjunctive Normal Form Theorem). *Every formula $\varphi \in \text{WFF}$ is *logically equivalent* to its *canonical conjunctive normal form* relative to a fixed ordering of its finitely many variables.*

$$\varphi \equiv \text{CCNF}_{P_1, \dots, P_n}(\varphi).$$

Go to proof.

Theorem 0.7.27 (Semantic Classification by Normal Forms). *Normal forms give finite certificates for semantic classification. For any formula $\varphi \in \text{WFF}$:*

1. φ is *satisfiable* if and only if some *DNF* formula equivalent to φ has a satisfiable conjunction term.
2. φ is a *contradiction* if and only if some *DNF* formula equivalent to φ has no satisfiable conjunction term.
3. φ is a *tautology* if and only if some *CNF* formula equivalent to φ has no falsifiable clause.
4. φ is a *contingency* if and only if it is satisfiable and not a tautology.

Go to proof.

Summary

0.8 Functional Completeness

Orientation

Boolean Functions and Representation

Definition 0.8.1 (Representation of a Boolean Function). Let $f : \mathbb{B}^n \rightarrow \mathbb{B}$ be a Boolean function. A formula $\varphi \in \text{WFF}$ with variables among $P_1, \dots, P_n$ *represents* f if for every truth assignment v ,

$$\widehat{v}(\varphi) = f(v(P_1), \dots, v(P_n)).$$

Theorem 0.8.2 (DNF Representation of Boolean Functions). *Every Boolean function $f : \mathbb{B}^n \rightarrow \mathbb{B}$ is represented by a formula in disjunctive normal form using variables $P_1, \dots, P_n$.*

Go to proof.

Connective Bases

Definition 0.8.3 (Connective Basis). A *connective basis* is a specified set B of propositional connectives allowed as primitive connectives for building formulas.

Definition 0.8.4 (Definable Connective). Let B be a connective basis. A connective $\circ$ is *definable from B* if there is a formula built using only connectives from B whose truth function agrees with the truth function of $\circ$.

Definition 0.8.5 (Functionally Complete Connective Set). A connective basis B is *functionally complete* if every Boolean function $f : \mathbb{B}^n \rightarrow \mathbb{B}$, for every $n \geq 1$, is represented by some formula using only connectives from B .

Definition 0.8.6 (Expressively Equivalent Connective Sets). Two connective bases B and C are *expressively equivalent* if every connective in B is definable from C , and every connective in C is definable from B .

Completeness of the Standard Connectives

Theorem 0.8.7 (Standard Connectives are Functionally Complete). *The connective basis*

$$\{\neg, \wedge, \vee\}$$

is functionally complete.

Go to proof.

Theorem 0.8.8 (Negation and Conjunction are Functionally Complete). *The connective basis*

$$\{\neg, \wedge\}$$

is functionally complete.

Go to proof.

Theorem 0.8.9 (Negation and Disjunction are Functionally Complete). *The connective basis*

$$\{\neg, \vee\}$$

is functionally complete.

Go to proof.

Reduction of Connective Sets

Definition 0.8.10 (Connective Reduction). A *connective reduction* replaces a formula using one connective basis by a logically equivalent formula using a smaller or different connective basis.

NAND

Definition 0.8.11 (NAND Connective). The *NAND* connective, written $\uparrow$, is the binary connective defined by

$$\varphi \uparrow \psi \equiv \neg(\varphi \wedge \psi).$$

Definition 0.8.12 (NAND-Only Formula). A *NAND-only formula* is a well-formed formula whose only connective is $\uparrow$.

Theorem 0.8.13 (NAND is Functionally Complete). *The one-connective basis $\{\uparrow\}$ is functionally complete. Go to proof.*

NOR

Definition 0.8.14 (NOR Connective). The *NOR* connective, written $\downarrow$, is the binary connective defined by

$$\varphi \downarrow \psi \equiv \neg(\varphi \vee \psi).$$

Definition 0.8.15 (NOR-Only Formula). A *NOR-only formula* is a well-formed formula whose only connective is $\downarrow$.

Theorem 0.8.16 (NOR is Functionally Complete). *The one-connective basis $\{\downarrow\}$ is functionally complete. Go to proof.*

Incomplete Connective Sets

Definition 0.8.17 (Incomplete Connective Set). A connective basis is *incomplete* if it is not functionally complete.

Definition 0.8.18 (Monotone Boolean Function). A Boolean function $f : \mathbb{B}^n \rightarrow \mathbb{B}$ is *monotone* if changing input coordinates from **False** to **True** never changes the output from **True** to **False**.

Theorem 0.8.19 (Conjunction and Disjunction are not Functionally Complete). *The connective basis $\{\wedge, \vee\}$ is not functionally complete.*

Go to proof.

Minimal Complete Bases

Definition 0.8.20 (Minimal Complete Basis). A functionally complete connective basis B is *minimal complete* if no proper subset of B is functionally complete.

Theorem 0.8.21 (NAND and NOR are Minimal Complete Bases). *The one-connective bases $\{\uparrow\}$ and $\{\downarrow\}$ are minimal complete bases.*

Go to proof.

Summary and Transition

0.9 Argument Forms and Informal Proof Patterns

Orientation

Argument Forms

Definition 0.9.1 (Argument Form). An *argument form* is a schematic pattern consisting of a finite list of premise formulas and one conclusion formula.

Definition 0.9.2 (Valid Argument Form). An argument form with premises Γ and conclusion φ is *valid* if every truth assignment that satisfies every formula in Γ also satisfies φ .

Definition 0.9.3 (Counterexample Assignment). A *counterexample assignment* to an argument form (Γ, φ) is a truth assignment that satisfies every premise in Γ and does not satisfy φ .

Theorem 0.9.4 (Truth-Table Validity Test for Argument Forms). *An argument form (Γ, φ) is valid if and only if no row of its truth table makes every formula in Γ true and φ false.*
Go to proof.

Core Valid Inference Patterns

Theorem 0.9.5 (Modus Ponens is Valid). *The argument form*

$$\varphi, \quad \varphi \rightarrow \psi \quad \therefore \quad \psi$$

is valid.

Go to proof.

Theorem 0.9.6 (Modus Tollens is Valid). *The argument form*

$$\varphi \rightarrow \psi, \quad \neg\psi \quad \therefore \quad \neg\varphi$$

is valid.

Go to proof.

Theorem 0.9.7 (Hypothetical Syllogism is Valid). *The argument form*

$$\varphi \rightarrow \psi, \quad \psi \rightarrow \chi \quad \therefore \quad \varphi \rightarrow \chi$$

is valid.

Go to proof.

Theorem 0.9.8 (Disjunctive Syllogism is Valid). *The argument form*

$$\varphi \vee \psi, \quad \neg\varphi \quad \therefore \quad \psi$$

is valid.

Go to proof.

Theorem 0.9.9 (Addition is Valid). *The argument form*

$$\varphi \quad \therefore \quad \varphi \vee \psi$$

is valid.

Go to proof.

Theorem 0.9.10 (Simplification is Valid). *The argument form*

$$\varphi \wedge \psi \quad \therefore \quad \varphi$$

is valid.

Go to proof.

Theorem 0.9.11 (Conjunction Introduction is Valid). *The argument form*

$$\varphi, \quad \psi \quad \therefore \quad \varphi \wedge \psi$$

is valid.

Go to proof.

Theorem 0.9.12 (Constructive Dilemma is Valid). *The argument form*

$$\varphi \rightarrow \chi, \quad \psi \rightarrow \theta, \quad \varphi \vee \psi \quad \therefore \quad \chi \vee \theta$$

is valid.

Go to proof.

Theorem 0.9.13 (Destructive Dilemma is Valid). *The argument form*

$$\varphi \rightarrow \chi, \quad \psi \rightarrow \theta, \quad \neg\chi \vee \neg\theta \quad \therefore \quad \neg\varphi \vee \neg\psi$$

is valid.

Go to proof.

Invalid Argument Forms

Theorem 0.9.14 (Affirming the Consequent is Invalid). *The argument form*

$$\varphi \rightarrow \psi, \quad \psi \quad \therefore \quad \varphi$$

is invalid.

Go to proof.

Theorem 0.9.15 (Denying the Antecedent is Invalid). *The argument form*

$$\varphi \rightarrow \psi, \quad \neg\varphi \quad \therefore \quad \neg\psi$$

is invalid.

Go to proof.

Rules of Replacement

Definition 0.9.16 (Rule of Replacement). A *rule of replacement* is a schematic permission to replace a subformula by a logically equivalent formula inside a formula context.

Theorem 0.9.17 (Replacement Rules Preserve Logical Equivalence). *If a formula ψ is obtained from a formula φ by finitely many applications of rules of replacement, then $\varphi \equiv \psi$.*

Go to proof.

Informal Proof Patterns

Definition 0.9.18 (Informal Proof Pattern). An *informal proof pattern* is a reusable truth-preserving strategy for organizing an ordinary mathematical proof.

Semantic Justification of Argument Forms

Theorem 0.9.19 (Tautological Implication Gives a Valid Argument Form). *Let $\Gamma = \{\gamma_1, \dots, \gamma_n\} \subseteq \text{WFF}$ and let $\varphi \in \text{WFF}$. If*

$$(\gamma_1 \wedge \dots \wedge \gamma_n) \rightarrow \varphi$$

is a tautology, then the argument form (Γ, φ) is valid.

Go to proof.

Argument Forms as a Bridge to Formal Systems

Summary and Transition

0.10 Compactness and Finite Satisfiability Preview

Orientation

Satisfiable Sets

Definition 0.10.1 (Satisfiable Set of Propositional Formulas). Let $\Gamma \subseteq \text{WFF}$. The set Γ is satisfiable if there exists a truth assignment v such that $v \models \Gamma$.

$$\text{SatisfiableSet}(\Gamma) \iff \exists v (\text{TruthAssignment}(v) \wedge v \models \Gamma).$$

Definition 0.10.2 (Unsatisfiable Set of Propositional Formulas). Let $\Gamma \subseteq \text{WFF}$. The set Γ is unsatisfiable if it is not satisfiable.

$$\text{UnsatisfiableSet}(\Gamma) \iff \neg \text{SatisfiableSet}(\Gamma).$$

Finitely Satisfiable Sets

Definition 0.10.3 (Finite Subset of a Set of Propositional Formulas). Let $\Gamma \subseteq \text{WFF}$. A finite subset of Γ is a set $\Delta \subseteq \Gamma$ containing only finitely many formulas.

Definition 0.10.4 (Finitely Satisfiable Set of Propositional Formulas). Let $\Gamma \subseteq \text{WFF}$. The set Γ is finitely satisfiable if every finite subset of Γ is satisfiable.

$$\text{FinitelySatisfiableSet}(\Gamma) \iff \forall \Delta ((\Delta \subseteq \Gamma \wedge \Delta \text{ is finite}) \Rightarrow \text{SatisfiableSet}(\Delta)).$$

Finite Unsatisfiability Witnesses

Definition 0.10.5 (Unsatisfiability Witness). Let $\Gamma \subseteq \text{WFF}$. A subset $\Delta \subseteq \Gamma$ is an unsatisfiability witness for Γ if Δ is unsatisfiable.

Definition 0.10.6 (Finite Unsatisfiability Witness). Let $\Gamma \subseteq \text{WFF}$. A subset $\Delta \subseteq \Gamma$ is a finite unsatisfiability witness for Γ if Δ is finite and unsatisfiable.

$$\text{FiniteUnsatisfiabilityWitness}(\Delta, \Gamma) \iff \Delta \subseteq \Gamma \wedge \Delta \text{ is finite} \wedge \text{UnsatisfiableSet}(\Delta).$$

Theorem 0.10.7 (Finite Unsatisfiability Witness). *Let $\Gamma \subseteq \text{WFF}$. If there exists a finite $\Delta \subseteq \Gamma$ such that Δ is unsatisfiable, then Γ is unsatisfiable.*

Go to proof.

Corollary 0.10.8 (Finite Witness Contrapositive). *Let $\Gamma \subseteq \text{WFF}$. If Γ is satisfiable, then every finite subset of Γ is satisfiable.*

Go to proof.

Propositional Compactness

Theorem 0.10.9 (Propositional Compactness). *Let $\Gamma \subseteq \text{WFF}$. If every finite subset of Γ is satisfiable, then Γ is satisfiable.*

Go to proof.

Finite Truth Table Intuition

Summary

Proofs

Predicate Logic

0.11 Model-Theoretic View

0.12 Axiom Schemas

0.13 Syntax

0.13.1 Syntax of First-Order Logic: Terms and Formulas

Definition 0.13.1 (Variable). For a first-order language $\mathcal{L}$, a *variable* is a symbol drawn from the distinguished variable stock of $\mathcal{L}$. Variables are usually written

$$x, y, z, x_0, x_1, x_2, \dots$$

Definition 0.13.2 (Constant Symbol). For a first-order language $\mathcal{L}$, a *constant symbol* is a non-logical symbol of arity 0. A constant symbol may occur as a term without taking input terms.

Definition 0.13.3 (Function Symbol). For a first-order language $\mathcal{L}$, an n -ary *function symbol* is a non-logical symbol of arity $n \geq 1$. If f has arity n , then it forms terms by the pattern

$$f(t_1, \dots, t_n).$$

Definition 0.13.4 (Predicate Symbol). For a first-order language $\mathcal{L}$, an n -ary *predicate symbol* is a non-logical symbol of arity $n \geq 1$. If P has arity n , then it forms atomic formulas by the pattern

$$P(t_1, \dots, t_n).$$

Definition 0.13.5 (Term). The set of *terms* of a first-order language $\mathcal{L}$, denoted $\text{Term}_{\mathcal{L}}$, is the smallest set of strings satisfying:

1. every variable is a term;
2. every constant symbol is a term;
3. if f is an n -ary function symbol and $t_1, \dots, t_n$ are terms, then $f(t_1, \dots, t_n)$ is a term;
4. no string is a term unless it is generated by finitely many applications of the previous clauses.

Definition 0.13.6 (Atomic Formula). An *atomic formula* of a first-order language $\mathcal{L}$ is a string of the form

$$P(t_1, \dots, t_n),$$

where P is an n -ary predicate symbol and $t_1, \dots, t_n$ are terms.

Definition 0.13.7 (Formula). For a first-order language $\mathcal{L}$, a *formula* is an expression formed by the formula grammar of $\mathcal{L}$. The grammar starts from atomic formulas and closes under the logical connectives and quantifiers of the language.

Definition 0.13.8 (Well-Formed Formula). The set of *well-formed formulas* of a first-order language $\mathcal{L}$, denoted $\text{Form}_{\mathcal{L}}$, is the smallest set of strings satisfying:

1. every atomic formula is a well-formed formula;
2. if φ is a well-formed formula, then $\neg\varphi$ is a well-formed formula;
3. if φ and ψ are well-formed formulas and $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then $(\varphi \circ \psi)$ is a well-formed formula;
4. if φ is a well-formed formula and x is a variable, then $\forall x \varphi$ and $\exists x \varphi$ are well-formed formulas;
5. no string is a well-formed formula unless it is generated by finitely many applications of the previous clauses.

Definition 0.13.9 (Molecular Formula). A *molecular formula* is a well-formed formula that is not atomic. Equivalently, it is a well-formed formula whose final formation step uses a logical connective or a quantifier.

0.13.2 Syntax: Free Variables, Bound Variables, and Substitution

Definition 0.13.10 (Scope of a Quantifier). Let φ be a formula of a first-order language.

If φ is of the form $\forall x \psi$ or $\exists x \psi$, then the formula ψ is called the *scope* of the quantifier.

The quantifier is said to *bind* all occurrences of the variable x that appear within its scope.

Definition 0.13.11 (Bound and Free Occurrences). An occurrence of a variable x in a formula φ is *bound* if it lies within the scope of a quantifier $\forall x$ or $\exists x$.

An occurrence of x is *free* if it is not bound by any quantifier in φ .

Definition 0.13.12 (Free Variables of a Formula). The set $\text{FV}(\varphi)$ of *free variables* of a formula φ is defined recursively:

1. If $\varphi = P(t_1, \dots, t_n)$ is atomic, then $\text{FV}(\varphi) = \bigcup_{i=1}^n \text{Var}(t_i)$, where $\text{Var}(t_i)$ is the set of variables in term t_i .
2. $\text{FV}(\neg\varphi) = \text{FV}(\varphi)$.
3. $\text{FV}(\varphi \circ \psi) = \text{FV}(\varphi) \cup \text{FV}(\psi)$ for any binary connective $\circ$.
4. $\text{FV}(\forall x \varphi) = \text{FV}(\varphi) \setminus \{x\}$.
5. $\text{FV}(\exists x \varphi) = \text{FV}(\varphi) \setminus \{x\}$.

Definition 0.13.13 (Sentence). A *sentence* (or *closed formula*) is a formula with no free variables.

Formally, φ is a sentence if and only if $\text{FV}(\varphi) = \emptyset$.

Definition 0.13.14 (Substitution Notation). Let φ be a formula, x a variable, and t a term.

The expression $\varphi[t/x]$ denotes the formula obtained from φ by replacing every *free* occurrence of x with the term t , leaving bound occurrences of x unchanged.

Definition 0.13.15 (Free for Substitution). A term t is *free for x* in φ if no free occurrence of x in φ lies within the scope of a quantifier $\forall y$ or $\exists y$ where y is a variable occurring in t .

Equivalently, t is free for x in φ if the substitution $\varphi[t/x]$ does not result in any variable in t becoming bound.

Definition 0.13.16 (Alpha-Equivalence). Formulas that differ only by the names of bound variables are logically equivalent. This is called *alpha-equivalence* (or *alphabetic variance*):

$$\forall x \varphi \equiv \forall y \varphi[y/x] \quad (\text{provided } y \text{ is not free in } \varphi).$$

0.13.3 Syntax: Formula Depth

Definition 0.13.17 (Formula Depth). The *depth* (or *complexity*) of a formula φ , denoted $\text{depth}(\varphi)$, is a natural number defined recursively:

1. **Atomic case.** If φ is atomic, then $\text{depth}(\varphi) = 0$.
2. **Negation.** If $\varphi = \neg\psi$, then $\text{depth}(\varphi) = \text{depth}(\psi) + 1$.
3. **Binary connectives.** If $\varphi = (\psi \circ \chi)$ for $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then
$$\text{depth}(\varphi) = \max\{\text{depth}(\psi), \text{depth}(\chi)\} + 1.$$
4. **Quantifiers.** If $\varphi = \forall x \psi$ or $\varphi = \exists x \psi$, then $\text{depth}(\varphi) = \text{depth}(\psi) + 1$.

0.14 Semantics

0.14.1 Semantics: Structures and Variable Assignments

Definition 0.14.1 (First-Order Language). A *first-order language* $\mathcal{L}$ consists of variables, logical symbols, and a signature of non-logical symbols: constant symbols, function symbols with assigned arities, and predicate symbols with assigned arities. Equality may be included as a distinguished logical predicate.

Definition 0.14.2 (Structure). Let $\mathcal{L}$ be a first-order language. A *structure* for $\mathcal{L}$ is a pair

$$\mathcal{M} = \langle D, I \rangle$$

where D is a nonempty set and I assigns each constant symbol an element of D , each n -ary function symbol a function $D^n \rightarrow D$, and each n -ary predicate symbol a relation on D^n .

Definition 0.14.3 (Variable Assignment). Let $\mathcal{M} = \langle D, I \rangle$ be a structure for $\mathcal{L}$. A *variable assignment* for $\mathcal{M}$ is a function

$$s : \text{Var}_{\mathcal{L}} \rightarrow D.$$

Definition 0.14.4 (Modified Assignment). Let $s : \text{Var}_{\mathcal{L}} \rightarrow D$, let $x \in \text{Var}_{\mathcal{L}}$, and let $d \in D$. The *modified assignment* $s[x \mapsto d]$ is the assignment defined by

$$s[x \mapsto d](y) = \begin{cases} d, & y = x, \\ s(y), & y \neq x. \end{cases}$$

Definition 0.14.5 (Term Evaluation). Let $\mathcal{M} = \langle D, I \rangle$ be a structure and s a variable assignment. The *value* of a term t , denoted $\llbracket t \rrbracket_{\mathcal{M}, s}$, is defined recursively by

$$\llbracket x \rrbracket_{\mathcal{M}, s} = s(x), \quad \llbracket c \rrbracket_{\mathcal{M}, s} = I(c),$$

and

$$\llbracket f(t_1, \dots, t_n) \rrbracket_{\mathcal{M}, s} = I(f)(\llbracket t_1 \rrbracket_{\mathcal{M}, s}, \dots, \llbracket t_n \rrbracket_{\mathcal{M}, s}).$$

0.14.2 Semantics: Satisfaction and Truth

Definition 0.14.6 (Satisfaction). Let $\mathcal{M} = \langle D, I \rangle$ be a structure, let s be a variable assignment, and let φ be a well-formed formula. The relation $\mathcal{M}, s \models \varphi$ is defined recursively by the usual clauses: atomic formulas are evaluated by term values and interpreted predicate relations; equality compares term values; connectives follow the truth conditions for $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$; and

$$\mathcal{M}, s \models \forall x \psi \iff \forall d \in D (\mathcal{M}, s[x \mapsto d] \models \psi),$$

$$\mathcal{M}, s \models \exists x \psi \iff \exists d \in D (\mathcal{M}, s[x \mapsto d] \models \psi).$$

Definition 0.14.7 (Truth in a Structure). Let φ be a sentence and let $\mathcal{M}$ be a structure. The sentence φ is *true in $\mathcal{M}$* , written $\mathcal{M} \models \varphi$, exactly when $\mathcal{M}, s \models \varphi$ for every variable assignment s for $\mathcal{M}$.

Definition 0.14.8 (Validity). A formula φ is *valid*, or *logically true*, exactly when $\mathcal{M}, s \models \varphi$ for every structure $\mathcal{M}$ for the language of φ and every variable assignment s for $\mathcal{M}$.

Definition 0.14.9 (Satisfiability). A formula φ is *satisfiable* exactly when there are a structure $\mathcal{M}$ and a variable assignment s such that $\mathcal{M}, s \models \varphi$.

0.14.3 Semantics: Models, Theories, and Logical Consequence

Definition 0.14.10 (Model of a Sentence). Let φ be a sentence and let $\mathcal{M}$ be a structure for the language of φ . The structure $\mathcal{M}$ is a *model of φ* exactly when $\mathcal{M} \models \varphi$.

Definition 0.14.11 (Model of a Set of Sentences). Let T be a set of sentences and let $\mathcal{M}$ be a structure for their common language. The structure $\mathcal{M}$ is a *model of T* , written $\mathcal{M} \models T$, exactly when $\mathcal{M} \models \theta$ for every $\theta \in T$.

Definition 0.14.12 (Theory). In Volume I, a *theory* is a set of sentences in a fixed first-order language.

Definition 0.14.13 (Logical Consequence). Let T be a set of sentences and let φ be a sentence in the same language. The sentence φ is a *logical consequence* of T , written $T \models \varphi$, exactly when every model of T is a model of φ .

Definition 0.14.14 (Logical Equivalence). Two sentences φ and ψ are *logically equivalent*, written $\varphi \equiv \psi$, exactly when each is a logical consequence of the other:

$$\varphi \models \psi \quad \text{and} \quad \psi \models \varphi.$$

0.14.4 Semantics: Open Formulas, Definable Relations, and Substitution Lemmas

Definition 0.14.15 (Open Formula). An *open formula* is a well-formed formula with at least one free variable. When the free variables are among $x_1, \dots, x_n$, it may be displayed as $\varphi(x_1, \dots, x_n)$.

Definition 0.14.16 (Definable Relation). Let $\mathcal{M} = \langle D, I \rangle$ be a structure and let $\varphi(x_1, \dots, x_n)$ be an open formula. The *relation defined by φ in $\mathcal{M}$* is the set

$$R_\varphi^\mathcal{M} = \{(a_1, \dots, a_n) \in D^n : \mathcal{M}, s[x_1 \mapsto a_1, \dots, x_n \mapsto a_n] \models \varphi\}.$$

Lemma 0.14.17 (Substitution Lemma for Terms). *Let $\mathcal{M}$ be a structure, let s be a variable assignment, let x be a variable, and let t and u be terms. Then*

$$\llbracket t[u/x] \rrbracket_{\mathcal{M}, s} = \llbracket t \rrbracket_{\mathcal{M}, s[x \mapsto \llbracket u \rrbracket_{\mathcal{M}, s}] }.$$

Go to proof.

Lemma 0.14.18 (Substitution Lemma for Formulas). *Let φ be a formula and let u be a term free for x in φ . For every structure $\mathcal{M}$ and assignment s ,*

$$\mathcal{M}, s \models \varphi[u/x] \iff \mathcal{M}, s[x \mapsto \llbracket u \rrbracket_{\mathcal{M}, s}] \models \varphi.$$

Go to proof.

0.15 Quantifiers

Basic Syntax and Semantics

Definition 0.15.1 (Universal Formula). A *universal formula* is a well-formed formula of the form

$$\forall x \varphi,$$

where x is a variable and φ is a well-formed formula.

Definition 0.15.2 (Existential Formula). An *existential formula* is a well-formed formula of the form

$$\exists x \varphi,$$

where x is a variable and φ is a well-formed formula.

Definition 0.15.3 (Bounded Quantifier). A *bounded quantifier* is an abbreviation that restricts a quantifier to a specified set or predicate:

$$\begin{aligned}\forall x \in A \varphi &:= \forall x (x \in A \rightarrow \varphi), \\ \exists x \in A \varphi &:= \exists x (x \in A \wedge \varphi).\end{aligned}$$

Definition 0.15.4 (Sentence). A *sentence* is a well-formed formula with no free variables.

Definition 0.15.5 (Open Formula). An *open formula* is a well-formed formula with at least one free variable.

Definition 0.15.6 (Quantifier Scope). In a formula $Qx \varphi$, where $Q \in \{\forall, \exists\}$, the *scope* of the quantifier Qx is the formula φ . Occurrences of x inside that scope are bound by the quantifier unless an inner quantifier for x intervenes.

Theorem 0.15.7 (Negation of Bounded Quantifiers). *Let A be a set and let φ be a formula. Then*

$$\neg(\forall x \in A \varphi) \equiv \exists x \in A \neg \varphi, \quad \neg(\exists x \in A \varphi) \equiv \forall x \in A \neg \varphi.$$

Go to proof.

Quantifier Laws

Theorem 0.15.8 (Quantifier Negation Laws). *For every formula φ ,*

$$\neg \forall x \varphi \equiv \exists x \neg \varphi, \quad \neg \exists x \varphi \equiv \forall x \neg \varphi.$$

Go to proof.

Theorem 0.15.9 (Double Negation and Quantifiers). *For every formula φ ,*

$$\neg \neg \forall x \varphi \equiv \forall x \varphi, \quad \neg \neg \exists x \varphi \equiv \exists x \varphi.$$

Go to proof.

Theorem 0.15.10 (Same-Type Quantifier Commutation). *Quantifiers of the same type commute:*

$$\forall x \forall y \varphi \equiv \forall y \forall x \varphi, \quad \exists x \exists y \varphi \equiv \exists y \exists x \varphi.$$

Go to proof.

Theorem 0.15.11 (Mixed Quantifiers Do Not Commute). *In general,*

$$\forall x \exists y \varphi \not\equiv \exists y \forall x \varphi.$$

Go to proof.

Theorem 0.15.12 (Valid Quantifier Distribution). *For all formulas φ, ψ ,*

$$\begin{aligned}\forall x (\varphi \wedge \psi) &\equiv (\forall x \varphi) \wedge (\forall x \psi), \\ \exists x (\varphi \vee \psi) &\equiv (\exists x \varphi) \vee (\exists x \psi).\end{aligned}$$

Go to proof.

Theorem 0.15.13 (Invalid Quantifier Distribution). *In general,*

$$\begin{aligned}\forall x (\varphi \vee \psi) &\not\equiv (\forall x \varphi) \vee (\forall x \psi), \\ \exists x (\varphi \wedge \psi) &\not\equiv (\exists x \varphi) \wedge (\exists x \psi).\end{aligned}$$

Go to proof.

Theorem 0.15.14 (Vacuous Quantification). *If x does not occur free in φ , then*

$$\forall x \varphi \equiv \varphi, \quad \exists x \varphi \equiv \varphi.$$

Go to proof.

Theorem 0.15.15 (Renaming Bound Variables). *If y is free for x in φ and no capture is introduced, then*

$$\forall x \varphi \equiv \forall y \varphi[y/x], \quad \exists x \varphi \equiv \exists y \varphi[y/x].$$

Go to proof.

Prenex Normal Form

Definition 0.15.16 (Quantifier Prefix). A *quantifier prefix* is a finite string

$$Q_1 x_1 Q_2 x_2 \cdots Q_n x_n,$$

where each $Q_i \in \{\forall, \exists\}$ and each x_i is a variable.

Definition 0.15.17 (Matrix). In a formula $Q_1 x_1 \cdots Q_n x_n \psi$, the *matrix* is the trailing formula ψ . For prenex normal form, the matrix is required to be quantifier-free.

Definition 0.15.18 (Prenex Formula). A *prenex formula* is a formula of the form

$$Q_1 x_1 Q_2 x_2 \cdots Q_n x_n \psi,$$

where $Q_1 x_1 \cdots Q_n x_n$ is a quantifier prefix and ψ is quantifier-free.

Definition 0.15.19 (Prenex Normal Form). A formula ψ is a *prenex normal form* of a formula φ exactly when ψ is prenex and $\varphi \equiv \psi$.

Theorem 0.15.20 (Existence of Prenex Normal Form). *Every first-order formula is logically equivalent to some formula in prenex normal form. Go to proof.*

Logical Strength and Quantifier Order

Definition 0.15.21 (Stronger Formula). A sentence A is *stronger than* a sentence B exactly when

$$A \models B \quad \text{and} \quad B \not\models A.$$

Definition 0.15.22 (Weaker Formula). A sentence B is *weaker than* a sentence A exactly when A is stronger than B .

Definition 0.15.23 (Quantifier Order). The *quantifier order* of a prenex formula is the ordered list of quantifiers in its prefix. A change of order is significant when it changes which witnesses may depend on earlier variables.

Theorem 0.15.24 (Universal Implies Existential). *Over nonempty domains,*

$$\forall x \varphi(x) \models \exists x \varphi(x).$$

Go to proof.

Theorem 0.15.25 (Existential Does Not Imply Universal). *In general,*

$$\exists x \varphi(x) \not\models \forall x \varphi(x).$$

Go to proof.

Theorem 0.15.26 (Uniform Witness Implies Dependent Witness). *For every formula $\varphi(x, y)$,*

$$\exists y \forall x \varphi(x, y) \models \forall x \exists y \varphi(x, y).$$

Go to proof.

Theorem 0.15.27 (Dependent Witness Need Not Be Uniform). *In general,*

$$\forall x \exists y \varphi(x, y) \not\models \exists y \forall x \varphi(x, y).$$

Go to proof.

0.16 Proof Theory

0.16.1 Proof Theory: Quantifier Inference Rules

Definition 0.16.1 (Universal Instantiation — UI). From a universally quantified statement, infer any instance obtained by substituting a term for the bound variable:

$$\forall x \varphi \Rightarrow \varphi[t/x].$$

Definition 0.16.2 (Universal Generalization — UG). From a formula, infer a universally quantified statement:

$$\varphi \Rightarrow \forall x \varphi.$$

Definition 0.16.3 (Existential Instantiation — EI). From an existentially quantified statement, infer an instance with a fresh constant (witness):

$$\exists x \varphi \Rightarrow \varphi[c/x].$$

Definition 0.16.4 (Existential Generalization — EG). From a formula containing a term, infer an existential statement:

$$\varphi[t/x] \Rightarrow \exists x \varphi.$$

0.16.2 Proof Strategies for Quantified Statements

Strategy 1: Proving a Universal $\forall x \varphi(x)$

Strategy 2: Proving an Existential $\exists x \varphi(x)$

Strategy 3: Using a Universal $\forall x \varphi(x)$

Strategy 4: Using an Existential $\exists x \varphi(x)$

The Central Asymmetry: Universal vs. Existential Goals

Nested Quantifiers: The $\forall\exists$ Pattern in Analysis

A Complete Worked Example

0.16.3 Proof Theory: Equality Rules

Definition 0.16.5 (Equality Introduction — Reflexivity). For any term t , one may infer $t = t$.

Definition 0.16.6 (Equality Elimination — Substitution of Equals). From $t_1 = t_2$ and $\varphi[t_1/x]$, one may infer $\varphi[t_2/x]$.

Theorem 0.16.7 (Symmetry of Equality). *From $t_1 = t_2$, one may infer $t_2 = t_1$. Go to proof.*

Theorem 0.16.8 (Transitivity of Equality). *From $t_1 = t_2$ and $t_2 = t_3$, one may infer $t_1 = t_3$. Go to proof.*

Theorem 0.16.9 (Term Substitution under Equality). *If $t_1 = t_2$, then for any function symbol f , $f(\dots, t_1, \dots) = f(\dots, t_2, \dots)$. Go to proof.*

Theorem 0.16.10 (Predicate Substitution under Equality). *If $t_1 = t_2$ and P is an n -ary predicate symbol, then $P(\dots, t_1, \dots) \Leftrightarrow P(\dots, t_2, \dots)$. Go to proof.*

0.16.4 Proof Theory: Soundness and Completeness

Definition 0.16.11 (Soundness). A proof system is *sound* if every provable formula is valid:

$$\Gamma \vdash \varphi \implies \Gamma \models \varphi.$$

Definition 0.16.12 (Completeness). A proof system is *complete* if every valid formula is provable:

$$\Gamma \models \varphi \implies \Gamma \vdash \varphi.$$

Theorem 0.16.13 (Soundness of First-Order Logic). *The standard inference rules for first-order logic (UI, UG, EI, EG, and the propositional rules) are sound: they preserve truth in all structures. If $\Gamma \vdash \varphi$, then $\Gamma \models \varphi$. Go to proof.*

0.17 Translation

English to Logic and Scope Ambiguity

Definition 0.17.1 (Translation Key). A *translation key* is a finite record that fixes a domain of discourse and assigns intended English readings to the constant symbols, predicate symbols, and relation symbols used in a translation.

Definition 0.17.2 (Domain of Discourse). The *domain of discourse* for a translation is the collection of objects over which the variables of the translated formulas range.

Definition 0.17.3 (Predicate and Relation Symbols in a Translation). In a translation key, a unary predicate symbol names a property of one object, and an n -ary relation symbol names a relation among n objects.

Definition 0.17.4 (Constants and Variables in a Translation). A *constant* names a fixed object from the translation key, while a *variable* marks a place bound by a quantifier or left open for later binding.

Proposition 0.17.5 (Universal Translation Pattern). *In a broad domain, an English statement of the form “All A are B ” is translated by*

$$\forall x (A(x) \rightarrow B(x)).$$

Go to proof.

Proposition 0.17.6 (Existential Translation Pattern). *In a broad domain, an English statement of the form “Some A are B ” is translated by*

$$\exists x (A(x) \wedge B(x)).$$

Go to proof.

Proposition 0.17.7 (Conditional Translation Pattern). *An English statement of the form “If A , then B ” is translated by*

$$A \rightarrow B,$$

with quantifiers placed according to the variables occurring in A and B .

Go to proof.

Proposition 0.17.8 (Negation Translation Pattern). *The negation of a translated quantified statement is obtained by negating the whole formula and then pushing the negation inward using the quantifier negation laws.*

Go to proof.

Definition 0.17.9 (Scope Ambiguity). A sentence has *scope ambiguity* when the same ordinary-language wording admits two or more translations with different quantifier nesting or different connective scope.

Square of Opposition

Definition 0.17.10 (Universal Affirmative Form). The *universal affirmative* categorical form, called A , translates “All A are B ” as

$$\forall x (A(x) \rightarrow B(x)).$$

Definition 0.17.11 (Universal Negative Form). The *universal negative* categorical form, called E , translates “No A are B ” as

$$\forall x (A(x) \rightarrow \neg B(x)).$$

Definition 0.17.12 (Particular Affirmative Form). The *particular affirmative* categorical form, called I , translates “Some A are B ” as

$$\exists x (A(x) \wedge B(x)).$$

Definition 0.17.13 (Particular Negative Form). The *particular negative* categorical form, called O , translates “Some A are not B ” as

$$\exists x (A(x) \wedge \neg B(x)).$$

Proposition 0.17.14 (A-O Contradiction). *The A -form $\forall x (A(x) \rightarrow B(x))$ and the O -form $\exists x (A(x) \wedge \neg B(x))$ are contradictory: exactly one of them is true.*

Go to proof.

Proposition 0.17.15 (E-I Contradiction). *The E -form $\forall x (A(x) \rightarrow \neg B(x))$ and the I -form $\exists x (A(x) \wedge B(x))$ are contradictory: exactly one of them is true.*

Go to proof.

Proposition 0.17.16 (Contrariety Requires Existential Import). *If $\exists x A(x)$, then the A -form and E -form cannot both be true.*

Go to proof.

0.18 Reference

Common Errors and Fallacies

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Languages, Axioms, and Models

Languages, Axioms, and Models Toolkit - Planned

0.32 Signatures

Signatures

Definition 0.32.1 (First-Order Signature). A *first-order signature* is a tuple

$$\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \text{ar}),$$

where $\mathcal{C}$ is a set of constant symbols, $\mathcal{F}$ is a set of function symbols, $\mathcal{R}$ is a set of relation symbols, and

$$\text{ar} : \mathcal{F} \cup \mathcal{R} \rightarrow \mathbb{Z}_{>0}$$

assigns to each function or relation symbol its number of arguments.

Definition 0.32.2 (Arity). Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \text{ar})$ be a first-order signature. The *arity function* is the map

$$\text{ar} : \mathcal{F} \cup \mathcal{R} \rightarrow \mathbb{Z}_{>0}$$

that assigns to each function or relation symbol the number of arguments it requires. When constants are treated as 0-ary function symbols, the codomain is enlarged to include 0.

Definition 0.32.3 (Language). A *first-order language* has two standard readings. As vocabulary, it is the same data as a signature,

$$\mathcal{L} = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \text{ar}).$$

As a formal language generated by a signature Σ , it is the set of well-formed strings built from the alphabet

$$A = \Sigma \cup V \cup \Lambda,$$

where V is a countable set of variables and Λ is the fixed stock of logical symbols and punctuation. In this second reading,

$$\mathcal{L}(\Sigma) = \{s \in A^* \mid s \in \text{Terms}(\Sigma) \text{ or } s \in \text{WFF}(\Sigma)\}.$$

Definition 0.32.4 (Constant Symbol). A *constant symbol* is a non-logical symbol that behaves syntactically as a 0-ary function symbol. In the streamlined convention where constants are included among function symbols, this is the condition

$$\text{Constant}(c) \iff c \in \mathcal{F} \text{ and } \text{ar}(c) = 0.$$

Because its arity is 0, the symbol c is already a well-formed term and takes no input terms.

Definition 0.32.5 (Function Symbol). A *function symbol* is a non-logical symbol $f \in \mathcal{F}$ whose assigned arity determines how many terms it must take as inputs. If $\text{ar}(f) = n$, then f is an n -ary term constructor: it takes exactly n terms and produces a new term, not a truth value.

Definition 0.32.6 (Relation Symbol). A *relation symbol*, or predicate symbol, is a non-logical symbol $R \in \mathcal{R}$ whose assigned arity determines how many terms it must take as inputs. If $\text{ar}(R) = n$, then R is an n -ary formula constructor: it takes exactly n terms and produces an atomic formula, hence something with a truth value rather than an object.

0.33 Axioms

Axioms

Definition 0.33.1 (Axiom). Let $\mathcal{L}$ be a first-order language, and let $\text{Sent}(\mathcal{L})$ denote the set of sentences of $\mathcal{L}$. If $T \subseteq \text{Sent}(\mathcal{L})$ is a theory, then an *axiom of T* is an element $\alpha \in T$.

Definition 0.33.2 (Axiom System). An *axiom system* is a collection of axioms expressed in a common language.

Definition 0.33.3 (Assumption). An *assumption* is a statement temporarily accepted for purposes of reasoning.

0.34 Theories

Theories

Definition 0.34.1 (Theory). A *theory* over a first-order language $\mathcal{L}$ is a collection of closed sentences in that language:

$$T \subseteq \text{Sentences}(\mathcal{L}).$$

If Γ is a chosen set of axioms, the theory generated by Γ is the deductive closure

$$T = \{\varphi \in \text{Sentences}(\mathcal{L}) \mid \Gamma \vdash \varphi\}.$$

Definition 0.34.2 (Theorem). A *theorem* of a theory T is a sentence φ for which there is a finite formal derivation from T , written

$$T \vdash \varphi.$$

Equivalently, in proof-theoretic terms, φ is a theorem of T exactly when some derivation has leaves drawn from logical axioms or sentences of T , uses valid inference rules, and has φ as its root.

Definition 0.34.3 (Consequence). A sentence φ is a *semantic consequence* of a set of premises Γ , written $\Gamma \models \varphi$, if every model of Γ is also a model of φ . Equivalently, there is no structure in which all sentences of Γ are true while φ is false.

Definition 0.34.4 (Deductive Closure). Let $\Gamma \subseteq \text{Sentences}(\mathcal{L})$ be a set of premises. The *deductive closure* of Γ , denoted $\text{Cn}(\Gamma)$, is the set of all sentences derivable from Γ :

$$\text{Cn}(\Gamma) = \{\varphi \in \text{Sentences}(\mathcal{L}) \mid \Gamma \vdash \varphi\}.$$

0.35 Models

Models

Definition 0.35.1 (Interpretation). Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \text{ar})$ be a first-order signature, and let M be a nonempty set. An *interpretation of Σ on M* is an assignment $\mathcal{I}$ that maps each symbol of Σ to concrete data over M : constants are assigned elements of M , function symbols are assigned operations on M , and relation symbols are assigned relations on M .

Definition 0.35.2 ($\mathcal{L}$ -Structure). Let $\mathcal{L}$ be a first-order signature. An *$\mathcal{L}$ -structure* is a pair

$$\mathfrak{M} = (M, \mathcal{I}),$$

where M is a nonempty domain and $\mathcal{I}$ is an interpretation of the symbols of $\mathcal{L}$ on M .

Definition 0.35.3 (Model). Let $\mathcal{L}$ be a first-order signature, let T be a theory over $\mathcal{L}$, and let $\mathfrak{M} = (M, \mathcal{I})$ be an $\mathcal{L}$ -structure. The structure $\mathfrak{M}$ is a *model of T* , written $\mathfrak{M} \models T$, if every sentence in T is true in $\mathfrak{M}$.

Definition 0.35.4 (Satisfaction). Let $\mathfrak{M} = (M, \mathcal{I})$ be an $\mathcal{L}$ -structure. A *variable assignment* is a function $s : \text{Var} \rightarrow M$. The assignment extends recursively to a term-evaluation map $\bar{s}$ by

$$\bar{s}(x) = s(x), \quad \bar{s}(c) = \mathcal{I}(c),$$

and

$$\bar{s}(f(t_1, \dots, t_n)) = \mathcal{I}(f)(\bar{s}(t_1), \dots, \bar{s}(t_n)).$$

The satisfaction relation $\mathfrak{M} \models \varphi[s]$ is then defined by induction on formulas: atomic relations are tested inside the interpreted relation, equality is interpreted as equality in M , Boolean connectives are interpreted by the corresponding truth operations, and quantifiers range over all elements of M .

Definition 0.35.5 (Countermodel). Let $\Gamma \subseteq \text{Sent}(\mathcal{L})$ be a set of premises, let $\varphi \in \text{Sent}(\mathcal{L})$, and let $\mathfrak{M}$ be an $\mathcal{L}$ -structure. A *countermodel to the entailment* $\Gamma \models \varphi$ is a structure $\mathfrak{M}$ that satisfies every sentence in Γ but does not satisfy φ .

0.36 Consistency

Consistency

Definition 0.36.1 (Contradiction). A *contradiction* is a statement that cannot be true under a given interpretation.

Definition 0.36.2 (Consistent Axiom System). A *consistent axiom system* is an axiom system that does not lead to contradiction.

Definition 0.36.3 (Inconsistent Axiom System). An *inconsistent axiom system* is an axiom system containing or producing contradictory assertions.

Definition 0.36.4 (Model Existence Criterion (Informal)). Informally, if an axiom system has a model, then the axioms are mutually compatible.

Definition 0.36.5 (Relative Consistency). Let T_0 and T_1 be theories. The theory T_1 is *consistent relative to* T_0 if

$$\text{Con}(T_0) \implies \text{Con}(T_1).$$

Equivalently, every contradiction in T_1 would imply a contradiction in T_0 .

0.37 Independence

Independence Toolkit - Planned

0.38 Comparing Theories

Comparing Theories Toolkit - Planned

0.39 Examples And Previews

Examples And Previews Toolkit - Planned

0.40 Summary

Summary Toolkit - Planned

Proofs